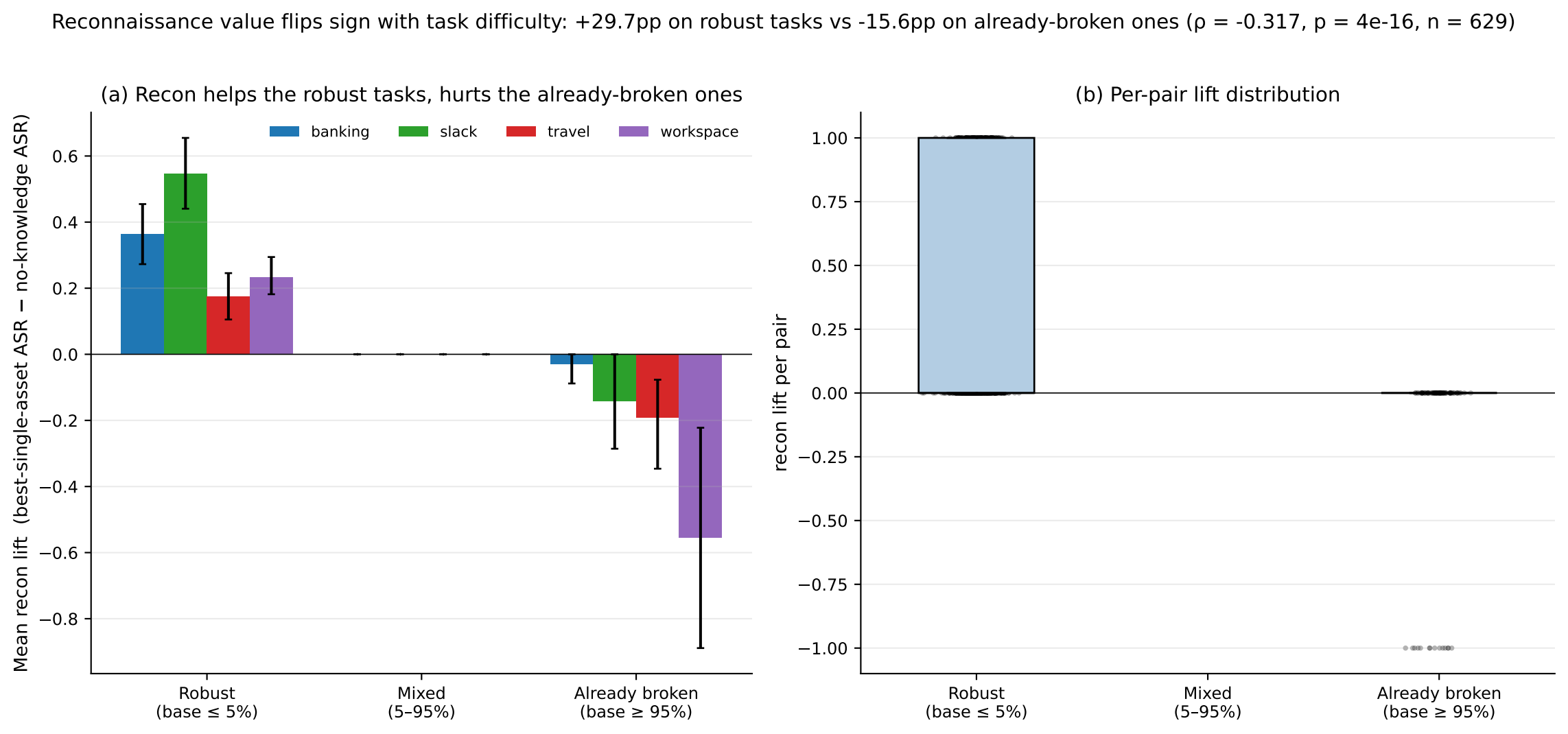
Reconnaissance value flips sign with task difficulty: +29.7pp on robust tasks vs -15.6pp on already-broken ones (ρ = -0.317, p = 4e-16, n = 629)
(a) Recon helps the robust tasks, hurts the already-broken ones
banking
slack
travel
workspace
Mean recon lift (best-single-asset ASR − no-knowledge ASR)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
Robust
(base ≤ 5%)
Mixed
(5–95%)
Already broken
(base ≥ 95%)
(b) Per-pair lift distribution
recon lift per pair
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Robust
(base ≤ 5%)
Mixed
(5–95%)
Already broken
(base ≥ 95%)